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GATE ECE · DIGITAL CIRCUITS · CHAPTER-WISE PYQS

Semiconductor Memories

GATE ECE Digital Circuits · Previous Year Questions with Answer Key

12	11	1992 – 2021	English
QUESTIONS	PAPERS COVERED	YEARS	LANGUAGE

INSIDE THIS PDF

Marks 1	7 questions
Marks 2	5 questions
Answer key	at the end

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Marks 1 7 questions

Q1. GATE ECE 2017 Set 2

In a DRAM,

(A) periodic refreshing is not required

(B) information is stored in a capacitor

(C) information is stored in a latch

(D) both read and write operations can be performed simultaneously

Open solution

Q2. GATE ECE 2015 Set 1

A 16 Kb (=16,384 bit) memory array is designed as a square with an aspect ratio of one (number of rows is equal to the number of columns). The minimum number of address lines needed for the row decoder is _____.

Open solution

Q3. GATE ECE 1996

Each cell of a static Random Access Memory Contains

(A) 6 MOS transistors.

(B) 4 MOS transistors and 2 capacitors

(C) 2 MOS transistors and 4 capacitors

(D) 1 MOS transistors and 1 capacitors

Open solution

Q4. GATE ECE 1995

The minimum number of MOS transistors required to make a dynamic RAM cell is

(A) 1 (B) 2 (C) 3 (D) 4

Open solution

Q5. GATE ECE 1994

A PLA can be used

(A) as a microprocessor

(B) as a dynamic memory

(C) to realize a sequential logic

(D) to realize a combinational logic

Open solution

Q6. GATE ECE 1994

A dynamic RAM consists of

(A) 6 transistors

(B) 2 transistors and 2 capacitor

(C) 1 transistors and 1 capacitor

(D) 2 capacitor only

Open solution

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Q7. GATE ECE 1992

Choose the correct satatement(s) from the following:

(A) PROM contains a programmable AND array and a fixed OR array.

(B) PLA contains a fixed AND array and a programmable OR array

(C) PROM contains a fixed AND array and a programmable OR array

(D) PLA contains a programmable AND array and a programmable OR array

Open solution

Marks 2 5 questions

Q8. GATE ECE 2021

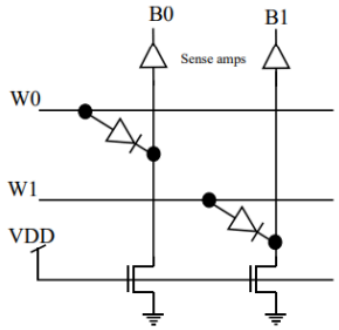
Addressing of a $32K \times 16$ memory is realized using a single decoder. The minimum number of AND gates required for the decoder is :

(A) 2^{19} (B) 2^{8} (C) 2^{32} (D) 2^{15}

Open solution

Q9. GATE ECE 2018

A 2×2 ROM array is built with the help of diodes as shown in the circuit below. Here W_0 and W_1 are signals that select the word lines and B_0 and B_1 are signals that are output of the sense amps based on the stored data corresponding to the bit lines during the read operation.



$$\begin{array}{c} & B_0 & B_1 \\ W_0 \\ W_1 \end{array} \begin{bmatrix} D_{00} & D_{01} \\ D_{10} & D_{11} \end{bmatrix}$$

Bits stored in the ROM Array

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During the read operation, the selected word line goes high and the other word line is in a high impedance state. As per the implementation shown in the circuit diagram above, what are the bits corresponding to D_{ij} (where i = 0 or 1 and j = 0 or 1) stored in the ROM?

(A) $\begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}$ (B) $\begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix}$

(C) $\begin{bmatrix} 1 & 0 \\ 1 & 0 \end{bmatrix}$ (D) $\begin{bmatrix} 1 & 1 \\ 0 & 0 \end{bmatrix}$

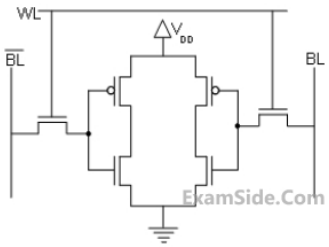
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Q10. GATE ECE 2014 Set 3

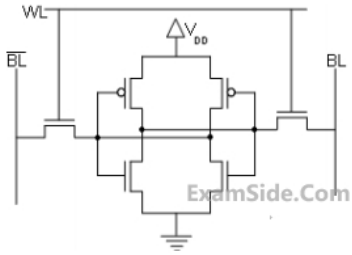
Open solution

If WL is the Word Line and BL the Bit Line, an SRAM cell is shown in

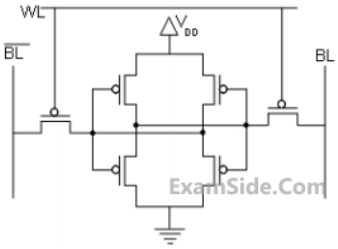
(A)



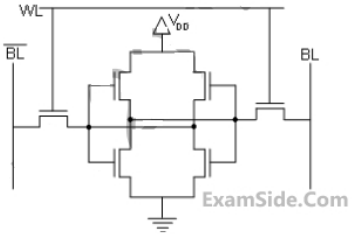
(B)



(C)



(D)



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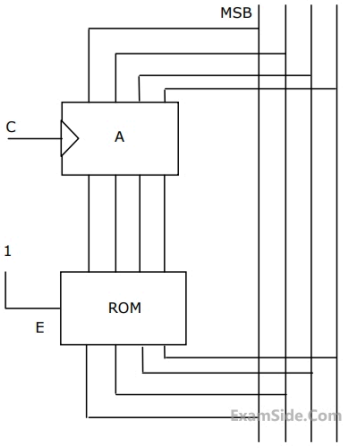
Q11. GATE ECE 2003

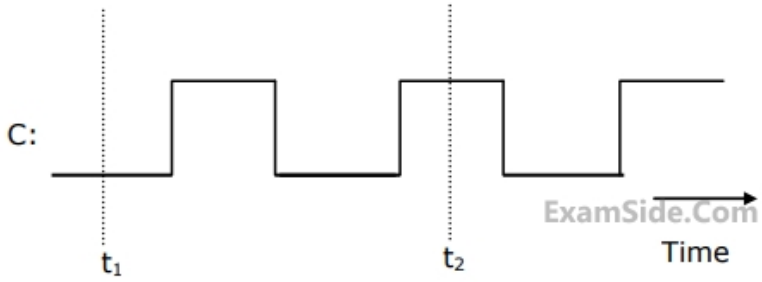
Open solution

In the circuit shown in Figure, A is a parallel in, parallel-out 4-bit register, which loads at the rising edge of the clock C. The input lines are connected to a 4-bit bus, W. Its output acts as the input to a 16×4 ROM whose output is floating when the enable input E is 0. A partial table of the contents of the ROM is as follows

Address	0	2	4	6	8	10	11	14
Data	0011	1111	0100	1010	1011	1000	0010	1000

The clock to the register is shown, and the data on the W bus at time t_1 is 0110. The data on the bus at time t_2 is





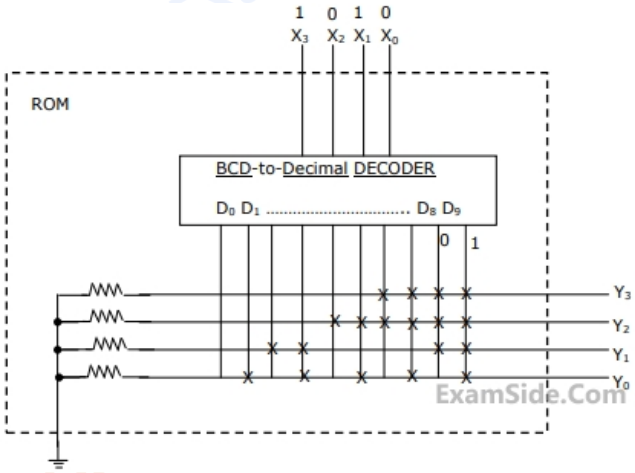
(A) 1111 (B) 1011 (C) 1000 (D) 0010

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Q12. GATE ECE 2002

Open solution

If the input X_3, X_2, X_1, X_0 to the ROM in figure 2.12 are 8-4-2-1 BCD numbers, then the outpus are Y_3,Y_2, Y_1, Y_0 are



(A) gray code numbers

(B) 2-4-2-1 BCD numbers

(C) excess-3 code numbeR

(D) none of the above.

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Answer Key

Marks 1

1. **B**	2. **7**	3. **A**	4. **A**	5. **D**
6. **C**	7. **C, D**			

Marks 2

8. **D**	9. **A**	10. **B**	11. **C**	12. **B**

For numerical answers given as a range, any value within the range is correct. Full solutions: scan the QR code next to each question, or visit questions.examside.com.

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